

Introduction:

In this programming assignment I was tasked to design and implement simple chatting app using TCP protocol with separate server and multiple clients. The requested design required us to use multithreaded approach to the design, and this design was implemented in my submission. Also please be aware that git bash terminal was used.

Methodology:

In this assignment I used python version 3.9.13, and I recommend use it as I don't know what is the oldest version that will be supported. In order to check my submission firstly unzip my archive into one folder and open 4 terminal and change their directories to that folder, after that firstly launch the server application with one the following command:

```
py Adil_Molzhigitov_server.py
```

```
python Adil_Molzhigitov_server.py
```

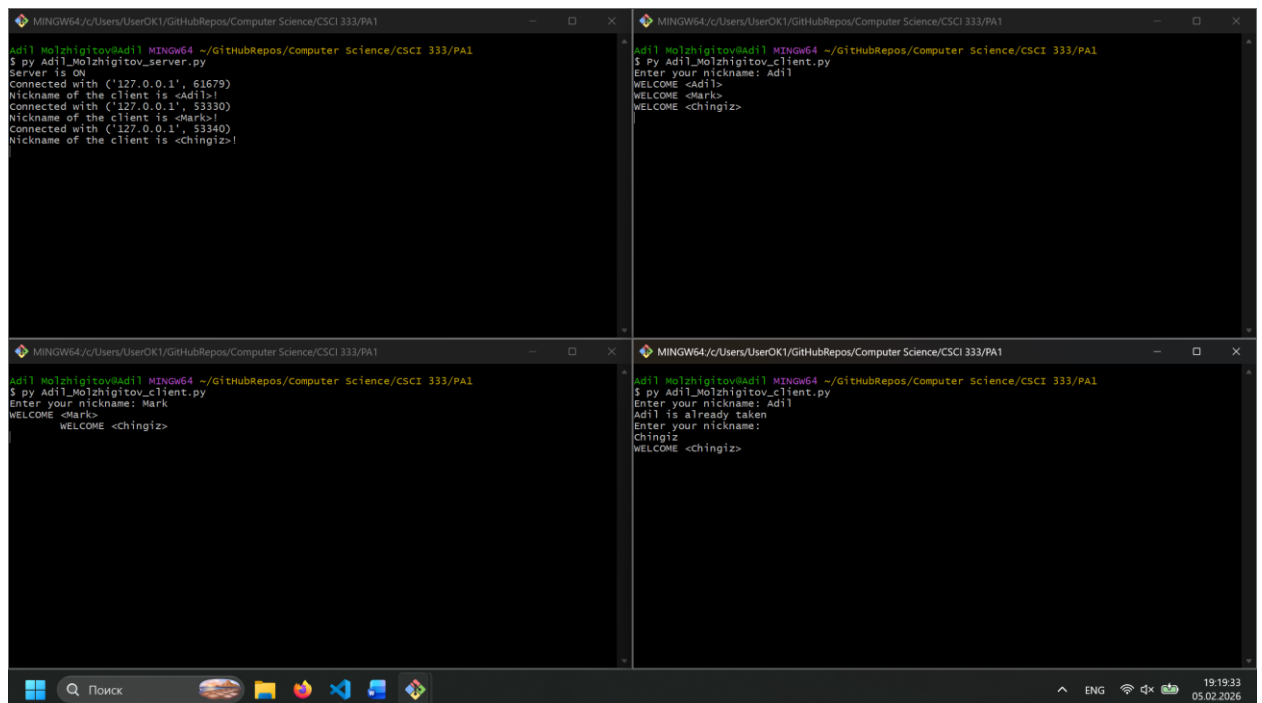
Then in separate terminals launch several (minimum 3 to check whole functionality) client applications with one of the following commands:

```
py Adil_Molzhigitov_client.py
```

```
python Adil_Molzhigitov_client.py
```

Below I will insert screenshots demonstrating completed functional requirements.

FR-1: (20%) Connection and Username



FR-2: (20%) Messaging

```
MINGW64/c/Users/UserOK1/GitHubRepos/Computer Science/CSCI 333/PA1
adil Molzhigitov@Adil MINGW64 ~/GitHubRepos/Computer Science/CSCI 333/PA1
$ py Adil_Molzhigitov_server.py
Server is ON
Connected with ('127.0.0.1', 61679)
Nickname of the client is <Adil>
Connected with ('127.0.0.1', 53330)
Nickname of the client is <Mark>
Connected with ('127.0.0.1', 53340)
Nickname of the client is <Chingiz>

MINGW64/c/Users/UserOK1/GitHubRepos/Computer Science/CSCI 333/PA1
adil Molzhigitov@Adil MINGW64 ~/GitHubRepos/Computer Science/CSCI 333/PA1
$ Py Adil_Molzhigitov_client.py
Enter your nickname: Adil
WELCOME <Adil>
WELCOME <Mark>
WELCOME <Chingiz>
MSG Hello everyone

MINGW64/c/Users/UserOK1/GitHubRepos/Computer Science/CSCI 333/PA1
adil Molzhigitov@Adil MINGW64 ~/GitHubRepos/Computer Science/CSCI 333/PA1
$ py Adil_Molzhigitov_client.py
Enter your nickname: Mark
WELCOME <Mark>
WELCOME <Chingiz>
FROM <Adil> <Hello everyone>

MINGW64/c/Users/UserOK1/GitHubRepos/Computer Science/CSCI 333/PA1
adil Molzhigitov@Adil MINGW64 ~/GitHubRepos/Computer Science/CSCI 333/PA1
$ py Adil_Molzhigitov_client.py
Enter your nickname: Adil
Adil is already taken
Enter your nickname:
Chingiz
WELCOME <Chingiz>
FROM <Adil> <Hello everyone>
```

FR-3: (20%) List Users

```
MINGW64/c/Users/UserOK1/GitHubRepos/Computer Science/CSCI 333/PA1
adil Molzhigitov@Adil MINGW64 ~/GitHubRepos/Computer Science/CSCI 333/PA1
$ Py Adil_Molzhigitov_client.py
Enter your nickname: Adil
WELCOME <Adil>
WELCOME <Mark>
WELCOME <Chingiz>
MSG Hello everyone
LIST
USERS <Adil> <Mark> <Chingiz>
```

FR-4: (20%) Private Message

```
MINGW64/c/Users/UserOK1/GitHubRepos/Computer Science/CSCI 333/PA1
adil Molzhigitov@adil MINGW64 ~/GitHubRepos/Computer Science/CSCI 333/PA1
$ py Adil_Molzhigitov_server.py
Server is ON
Connected with ('127.0.0.1', 61679)
Nickname of the client is <Adil>
Connected with ('127.0.0.1', 53330)
Nickname of the client is <Mark>
Connected with ('127.0.0.1', 53340)
Nickname of the client is <Chingiz>

adil Molzhigitov@adil MINGW64 ~/GitHubRepos/Computer Science/CSCI 333/PA1
$ py Adil_Molzhigitov_client.py
Enter your nickname: Adil
WELCOME <Adil>
WELCOME <Mark>
WELCOME <Chingiz>
MSG Hello everyone
LIST
USERS <Adil> <Mark> <Chingiz>
PM Mark private message for Mark

adil Molzhigitov@adil MINGW64 ~/GitHubRepos/Computer Science/CSCI 333/PA1
$ py Adil_Molzhigitov_client.py
Enter your nickname: Adil
Adil is already taken
Enter your nickname:
Chingiz
WELCOME <Chingiz>
FROM <Adil> <Hello everyone>

adil Molzhigitov@adil MINGW64 ~/GitHubRepos/Computer Science/CSCI 333/PA1
$ py Adil_Molzhigitov_client.py
Enter your nickname: Mark
WELCOME <Mark>
WELCOME <Chingiz>
FROM <Adil> <Hello everyone>
LIST
USERS <Adil> <Mark> <Chingiz>
PMFROM <Adil> <private message for Mark>
```

```
MINGW64/c/Users/UserOK1/GitHubRepos/Computer Science/CSCI 333/PA1

adil Molzhigitov@adil MINGW64 ~/GitHubRepos/Computer Science/CSCI 333/PA1
$ py Adil_Molzhigitov_client.py
Enter your nickname: Adil
Adil is already taken
Enter your nickname:
Chingiz
WELCOME <Chingiz>
FROM <Adil> <Hello everyone>
PM Nobody This is private message for Nobody
Error NoSuchUser
|
```

FR-5: (20%) Graceful Exit

```
MINGW64/c/Users/UserOK1/GitHubRepos/Computer Science/CSCI 333/PA1
adil Molzhigitov@adil MINGW64 ~/GitHubRepos/Computer Science/CSCI 333/PA1
$ py Adil_Molzhigitov_server.py
Server is ON
Connected with ('127.0.0.1', 61679)
Nickname of the client is <Adil>
Connected with ('127.0.0.1', 53330)
Nickname of the client is <Mark>
Connected with ('127.0.0.1', 53340)
Nickname of the client is <Chingiz>
Client <Adil> disconnected!
Client <Chingiz> disconnected!
Client <Mark> disconnected!

adil Molzhigitov@adil MINGW64 ~/GitHubRepos/Computer Science/CSCI 333/PA1
$ py Adil_Molzhigitov_client.py
Enter your nickname: Adil
WELCOME <Adil>
WELCOME <Mark>
WELCOME <Chingiz>
MSG Hello everyone
LIST
USERS <Adil> <Mark> <Chingiz>
PM Mark private message for Mark
QUIT
BYE

adil Molzhigitov@adil MINGW64 ~/GitHubRepos/Computer Science/CSCI 333/PA1
$ py Adil_Molzhigitov_client.py
Enter your nickname: Adil
Adil is already taken
Enter your nickname:
Chingiz
WELCOME <Chingiz>
FROM <Adil> <Hello everyone>
PM Nobody This is private message for Nobody
Error: NoSuchUser
<Adil> left the chat!
QUIT
BYE

adil Molzhigitov@adil MINGW64 ~/GitHubRepos/Computer Science/CSCI 333/PA1
$
```

Discussion:

In this PA I learned how using python implemented TCP connection using socket library and how to design multithreaded application using python's threading library. I learned that what is IP and PORT and that their combination gives me SOCKET. I also learned that in such application server used separate thread for each client. Also I learned that client has 2 threads where 1 listens for the messages and 2 used to write user input and send messages.